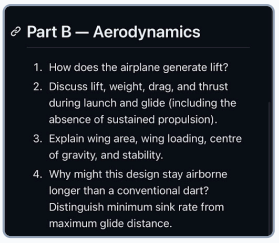
PART B — AERODYNAMICS STUDY GUIDE

The "Sky King" Paper Airplane: Why Every Fold Works

A simple, step-by-step physics explanation linking folding choices to flight performance

What makes the Sky King special? Designed by Takuo Toda, the *Sky King* broke the Guinness World Record for longest flight time (27.9 seconds in 2009, later 29.2 seconds) using a single uncut A4 sheet. Instead of flying fast like a spear, every fold in the Sky King is engineered to turn a flat piece of paper into a slow-sinking glider.



Why Do We Make Each Fold? (Fold-by-Fold Guide)

Sky King Fold	What It Physically Does	Aerodynamic Reason ("Why This Fold?")
1. Repeated Front Corner Folds (Steps 2–7)	Stacks 6 to 8 layers of paper tightly at the front of the sheet, leaving the back single-layered.	Moves the **Centre of Gravity (CG)** forward so the plane glides nose-first instead of flipping backward and stalling.
2. Blunt Square / Diamond Lock (Steps 10–14)	Folds the nose tip up and flattens it into a wide diamond pocket that clamps the underside shut.	Locks all front layers together without tape so the wings cannot burst open during a hard, high-speed upward throw.
3. Shallow Body Crease & Wide Wings (Step 15)	Uses only a narrow center handle for the body, leaving most of the paper spread out as wide wings.	Maximizes **Wing Area** and minimizes **Wing Loading**, letting the plane float on a large cushion of air.
4. Vertical Wingtips / Winglets (Step 16)	Folds the outer 1.75 cm of both wingtips straight upward at a 90° angle.	Blocks high-pressure air underneath from spilling over the wingtips, cutting drag and preventing sideways slipping.
5. Upward Back-Edge Curve / "m" Flex	Gently curls the rear edge of the wings upward and sets the wings in a slight upward "V" angle.	Keeps the nose slightly raised to generate **Lift** and automatically levels the wings if the plane tilts.

1. How Does the Airplane Generate Lift?

Lift is the upward push of air that holds the Sky King up against gravity. Because a paper plane is made from a flat sheet, its folds help generate lift in two simple ways as it moves forward through the air:

- **Pushing Air Downward (Angle of Attack & Action–Reaction):** By gently curling the back edge of the wings upward at the end of folding, the tail is pushed down slightly so the nose points up at a small angle (called the *angle of attack*). As the Sky King glides forward, the flat underside of its huge wings collides with oncoming air and deflects that air downward. By Newton's Third Law, pushing air down pushes the wings up.
- **Pressure Difference Across the Wing (Curved Airflow):** Because the front edge of the wing is folded over the thick multi-layer nose pocket while the back of the wing is a single thin layer, the top surface forms a gentle arch (*camber*). Air flowing over the curved top surface speeds up and creates slightly

2. Lift, Weight, Drag, and Thrust: Launch vs. Glide

Four forces act on the Sky King: **Thrust** (forward push), **Drag** (air resistance slowing it down), **Lift** (upward push from the wings), and **Weight** (gravity pulling it down). Because a paper airplane has **no engine (no sustained propulsion)**, these forces behave completely differently during launch versus glide:

- **Phase A — The Launch (Climbing Upward):** To stay aloft as long as possible, you throw the Sky King hard and nearly straight up. Your arm supplies a huge burst of initial **Thrust**. The moment the plane leaves your fingers, **Thrust becomes zero**. It coasts upward using its momentum while both **Weight** and **Drag** pull backward to slow it down. Here, the *Diamond Pocket Lock (Steps 10–14)* is critical: it holds the folded seams tight against intense launch wind so the wings stay tucked until the plane reaches peak height.
- **Phase B — The Transition & Unpowered Glide:** At the very top of the throw, the plane stops rising. Because of the heavy folded nose, the front tips downward smoothly instead of stalling backward. With no engine to create thrust, **gravity (Weight) acts as the engine**: by gliding down a gentle invisible ramp (glide slope), a small portion of gravity pulls the plane forward ($\mathbf{W \times \sin \theta}$). That forward motion keeps air flowing over the wide wings so that upward **Lift** and backward **Drag** balance the downward pull of **Weight** at a slow, steady speed.

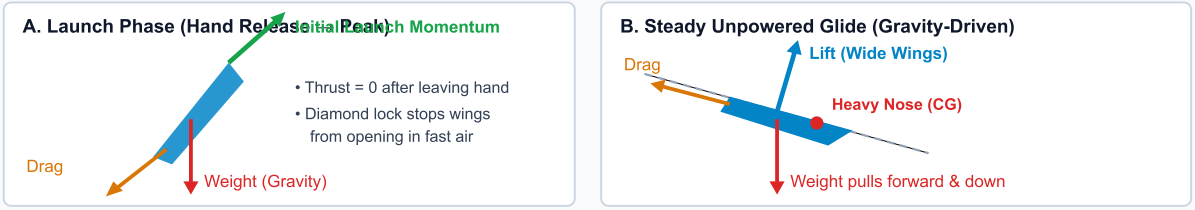


Figure 1: During launch (A), the diamond nose lock resists air drag. During glide (B), there is no engine thrust; a forward Centre of Gravity (CG) tilts the nose down slightly so gravity pulls the plane forward along a shallow slope, generating Lift.

3. Wing Area, Wing Loading, Centre of Gravity & Stability

- **Wing Area (How Big the Wings Are):** In Step 15, you fold a very shallow center body seam so almost the entire A4 sheet becomes wing surface. More wing area means the plane touches and pushes on a much larger volume of air at once.
- **Wing Loading ($\mathbf{Weight \div Wing\ Area}$):** Wing loading measures how much weight each square centimetre of wing has to carry. Because the Sky King uses the same single sheet of paper (~4.5 grams) as any other plane but spreads it over a **much larger wing area**, its wing loading is **extremely low**. Low wing loading allows it to fly at very slow speeds without sinking fast.
- **Centre of Gravity — CG (The Balance Point):** An unfolded sheet of paper balances in the middle, which causes it to flutter and flip backward when thrown. Folding the front corners inward 6 to 8 times (Steps 2–14) packs extra paper weight into the nose. This places the **CG slightly ahead of the wings' Centre of Lift**. Whenever the plane slows down, the heavier nose gently drops to pick up speed automatically.
- **Stability (Flying Straight Without Tumbling): (1) Pitch stability (up/down)** comes from the combination of the heavy folded nose (pulling the nose down) and the upward-curved rear wing edge (pushing the tail down), locking the glider at a steady angle. **(2) Roll & Yaw stability (side-to-side)** comes from folding the wings in a slight upward "V" (*dihedral*) and folding up the **1.75 cm winglets** (Step 16). If a gust tips one wing down, that lower wing meets more air pressure and pushes itself back

4. Why Sky King Stays Airborne Longer Than a Conventional Dart

A **conventional dart** has a sharp pointed nose and narrow triangular wings folded several times down the middle. While those narrow folds make a dart stiff and fast so it shoots far across a room, they eat up most of the paper's surface area. With small wings carrying the full weight of the paper (*high wing loading*), a dart must fly very fast just to generate enough lift—and therefore hits the ground in only 2 to 4 seconds.

The **Sky King** is folded for the exact opposite goal. Its wide, Blunt Nose fold leaves the back of the sheet wide open, creating giant wings with *low wing loading*, while its upturned **winglets** stop air from leaking around the wingtips (reducing *induced drag*). Because each square centimetre of wing carries tiny weight, the Sky King generates full lift even while moving at a crawl, allowing it to float slowly downward for nearly 30 seconds.

Design Feature	Sky King (Time-Aloft Glider)	Conventional Dart (Distance Flyer)
Nose & Body Folds	Blunt folded diamond nose; shallow center body seam saves paper for wings.	Long pointed nose; deep center folds put multiple heavy layers into the body.
Wing Area & Shape	**Very Large**, wide rectangular/swept wings with upright 1.75 cm winglets.	**Small**, narrow delta (triangle) wings with no winglets.
Wing Loading	**Very Low** (**Weight ÷ Large Area**) — floats lightly like a parachute or hawk.	**High** (**Weight ÷ Small Area**) — must fly fast like a spear to stay up.
Wingtip Air Leaks	Blocked by vertical winglets, saving energy at slow gliding speeds.	Unblocked; swirling wingtip vortices drain energy when slowing down.
Main Flight Strength	**Minimum Sink Rate** (stays in the air for the longest time in seconds).	**High Speed & Straight Distance** (covers ground quickly before landing).

Minimum Sink Rate vs. Maximum Glide Distance

In aerodynamics, "staying in the air the longest time" and "flying the farthest distance across the floor" are two different goals that require different speeds and wing folds:

- **Minimum Sink Rate (Longest Time Aloft — Sky King's Goal):** *Sink rate* is the vertical speed at which the plane drops toward the floor (measured in metres per second downward, $\mathbf{v_{sink}}$). To stay airborne for the maximum number of seconds, you only care about making $\mathbf{v_{sink}}$ as tiny as possible, even if the plane barely moves forward across the room. The Sky King's huge wing area, low wing loading, and upturned rear wing edges slow its forward speed down to the exact sweet spot where vertical falling speed is at an absolute minimum.
- **Maximum Glide Distance (Farthest Horizontal Reach):** *Glide distance* measures how many metres forward the plane travels for every metre of height it loses (determined by the **Lift-to-Drag ratio**, $\mathbf{L \div D}$, which equals $\mathbf{Forward\ Speed \div Sink\ Rate}$). To cover big horizontal distance before touching the ground, a plane must fly much faster forward with a sleeker, lower-drag shape (like a dart)—meaning it travels farther across the gym floor, but reaches the floor in far fewer seconds.

> **Simple Takeaway to Remember:** Think of **Maximum Glide Distance** like a fast zip-line that shoots you far across a field in 4 seconds, and **Minimum Sink Rate (Sky King)** like a wide parachute canopy that drifts gently straight down for 28 seconds. Every fold of the Sky King—heavy blunt nose, wide wings, and winglets—is built to minimize sink rate.